# Structural Design Report

N1 / G+1 Storey Commercial

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## Abstract

G+1 story low rise RCC structure is *Designed & Drafted* using *BNBC2020*. Various supporting simulation and drafting tools are used.

## 1 Project Information

- **Project:** G+1 story in Gopalganj
- **Location:** Gopalganj
- **Building Use:** Commercial
- **Structural System:** RCC frame

## 2 Design Basis

### 2.1 Codes & Standards

- BNBC 2020 (Loads, Concrete/Steel)
- ASCE, ACI (Design Load Combination)

### 2.2 Properties

**Table 1:** Material Properties

| Item | Symbol | Value |
|---|---|---|
| Concrete strength | $f'_c$ | 2500 psi |
| Steel yield | $f_y$ | 725 000 psi |
| Unit weight (RCC) | $\gamma_c$ | 150 lb/ft$^3$ |
| Brick Unit Weight | $\gamma_b$ | 120 lb/ft$^3$ |

## 3 Loads & Load Combinations

### 3.1 Gravity Loads

Loads that are used as gravity load are.

**Table 2:** Gravity load on zones

| Zone | Load |
|---|---|
| GF Top Slab | SW, LL_52, LL_RW, SDL_FF_25, SDL_FPW |
| FF Top Slab | SW, RLL_20, SDL_FF_25 |

**Table 3:** Loads

| Item | Symbol | Value |
|---|---|---|
| Load | $SW$ | calc |
| Load | $FF$ | calc or 25 psf |
| Load | $RW$ | calc or 25 psf |
| Load | $LL_{commercial}$ | 52 psf |
| Load | $RLL_{roof}$ | 20 psf |
| Load | $FPW_{int}$ | calculated 438 plf |

### 3.2 Lateral Loads

- Wind: Only in detailing.

**Table 4:** Seismic Parameters

| Item | Symbol | Value |
|---|---|---|
| Seismic Zone (Gopalganj) | $Zone$ | 1 |
| Seismic Zone Coefficient | $Z$ | 0.12 |
| Structure Importance Factor | $I$ | 1.0 |
| Response Modification Coefficient | $R$ | 5.0 |
| System Overstrength Factor | $\Omega_0$ | 3.0 |
| Deflection Amplification Factor | $C_d$ | 4.5 |
| Mapped Spectral Response (Short) | $S_s$ | 0.30 |
| Mapped Spectral Response (1 sec) | $S_1$ | 0.12 |
| Site Coefficient (Short Period)* | $F_a$ | 1.15 |
| Site Coefficient (1 sec Period)* | $F_v$ | 1.725 |
| Approximate Period (20ft height) | $T_a$ | 0.24 sec |

### 3.3 Load Combinations

$$1.2 * DeadLoad + 1.6 * LiveLoad$$

## 4 Structural Model

Structural 3D model is shown in Figure 1.

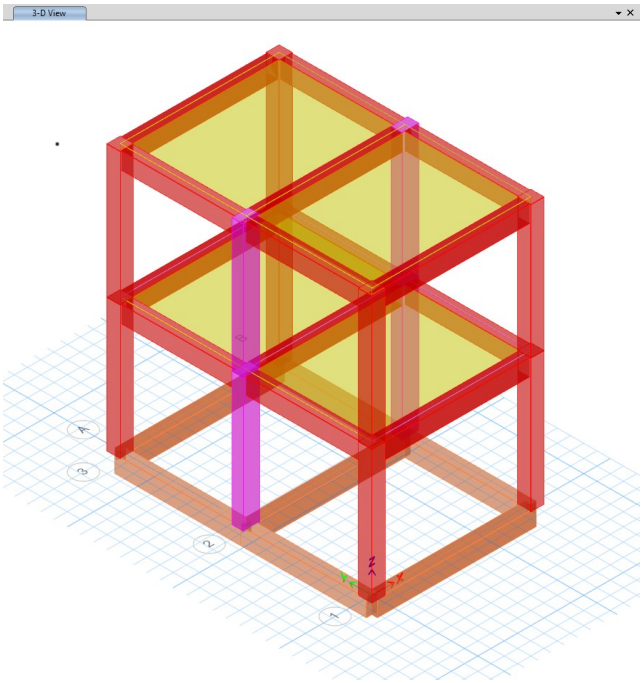


**Figure 1:** 3d view

## 5 Structural Analysis and Design

### 5.1 Slab

Figure 2 shows 1 directional rebar and thickness adequacy of GF level slabs. Similarly other slabs are adequate in both

direction.

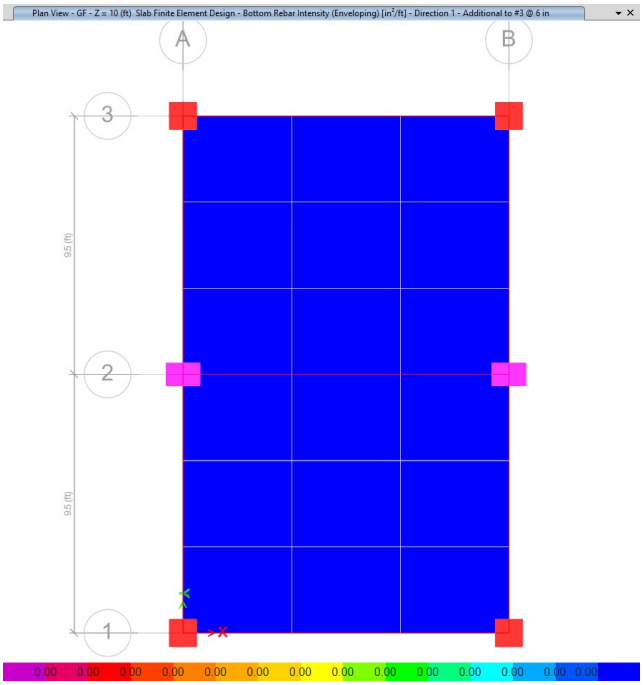


**Figure 2:** Slab thickness and rebar adequacy check.

### 5.1.1 Beam

Rebar area required square inch for GF level beams are showen in Figure 3. All the beams are designed and detailed accordingly.

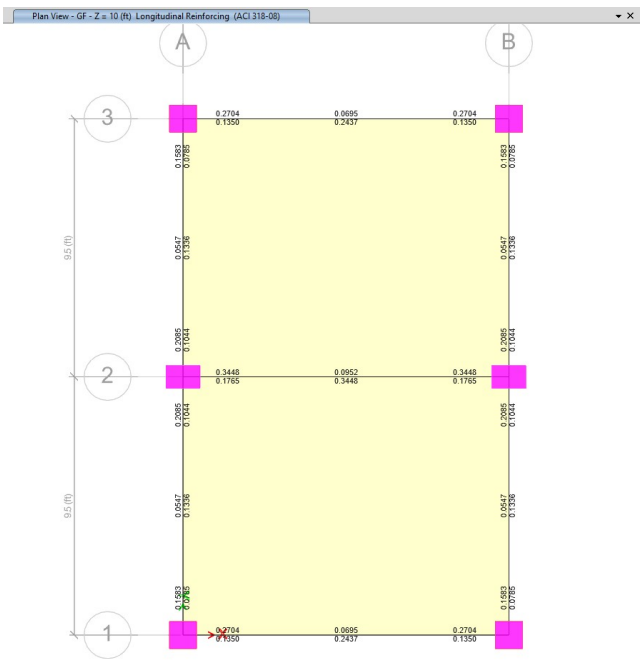


**Figure 3:** Rebar requirements for GF level Beams.

### 5.1.2 Beam

Rebar area required in percentage for one elevation view for columns are showen in Figure 4. All the columns required the minimum rebar area.

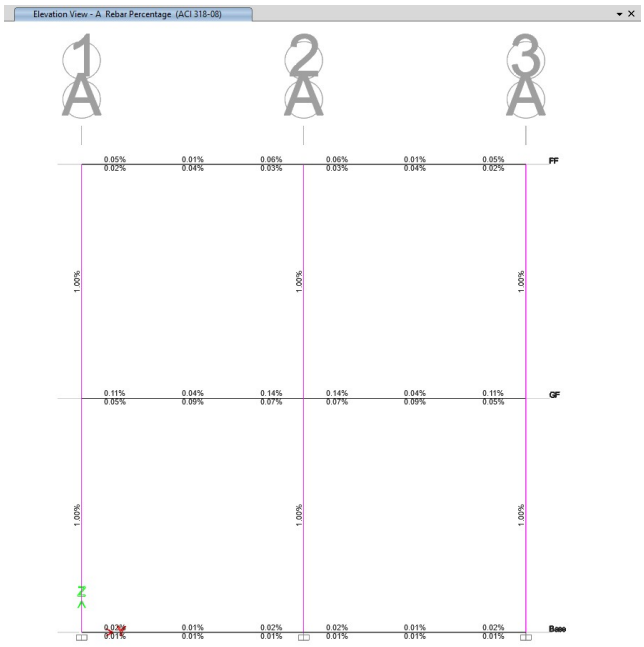


**Figure 4:** Rebar percentage requirements column at Elevation 3

### 5.1.3 Base reaction

Base reaction for the combined load of Dead and Live are shown in Figure 5. Using 1.1*(DL+LL) the footing size of columns are calculated and Using 1.2DL + 1.6LL load combination the footing thickness and their corresponding rebar detailing are done.

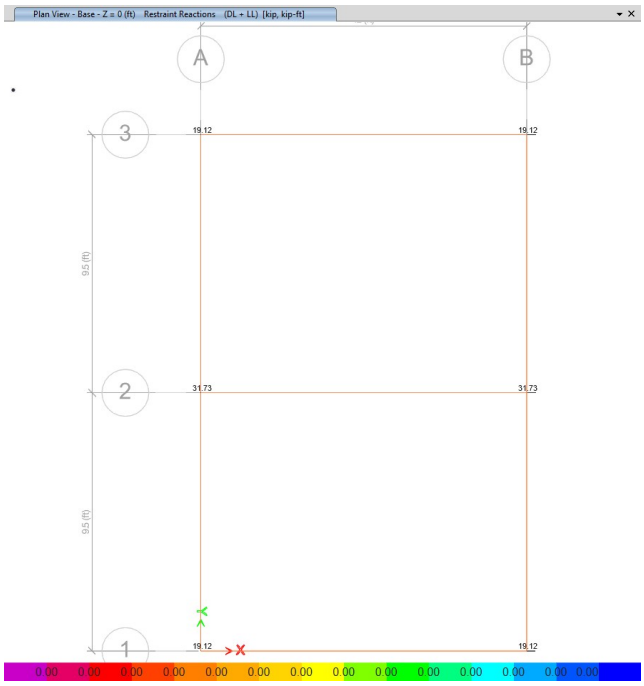


**Figure 5:** Base reactions considering Dead + Live load